

2.3.4 Normal (Gaussian)distribution

Random variable X has *normal (Gaussian) distribution* if density function is defined by the dependence:

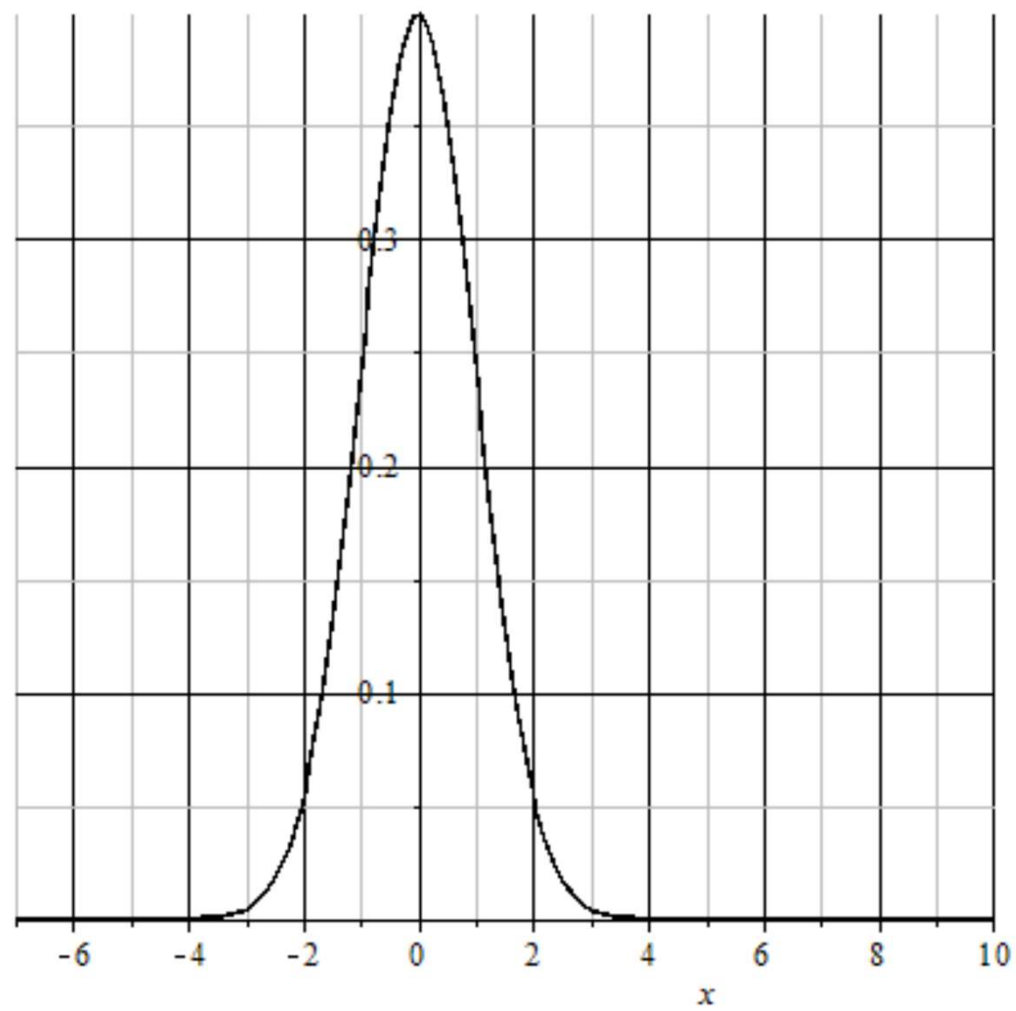
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu_X)^2}{2\sigma^2}}. \quad (32)$$

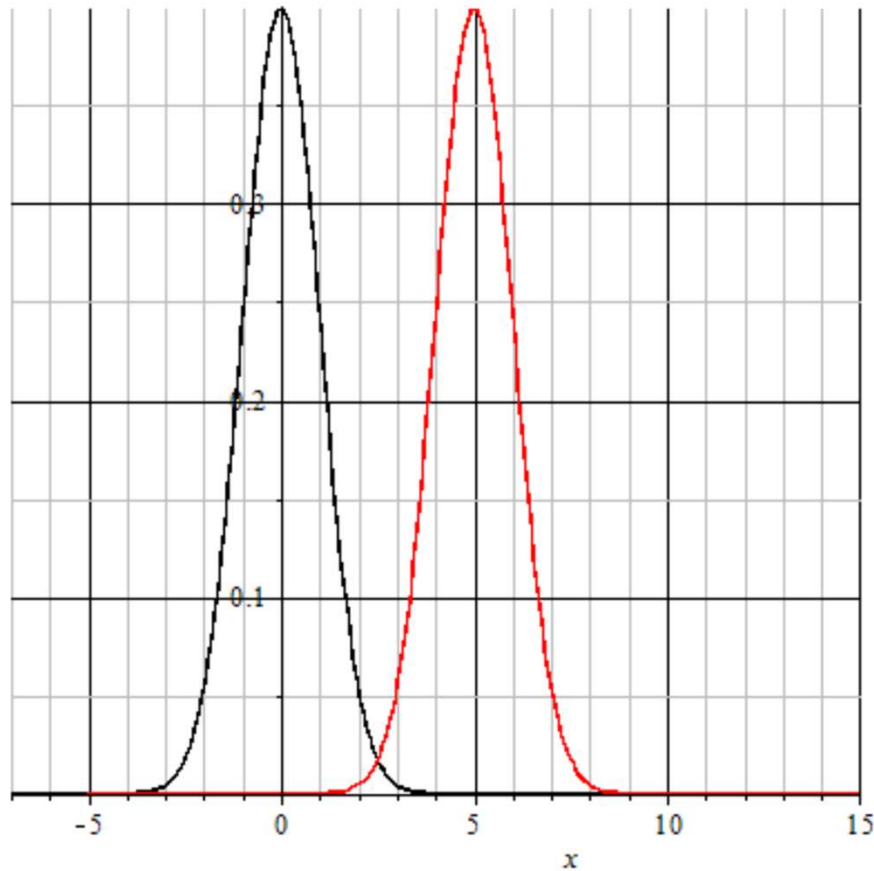
where *Mathematical expectation*: $E(X) = \mu_X$,

Standard deviation: σ

AT $\mu_X = 0, \sigma = 1$ normal distribution is called *standard*.

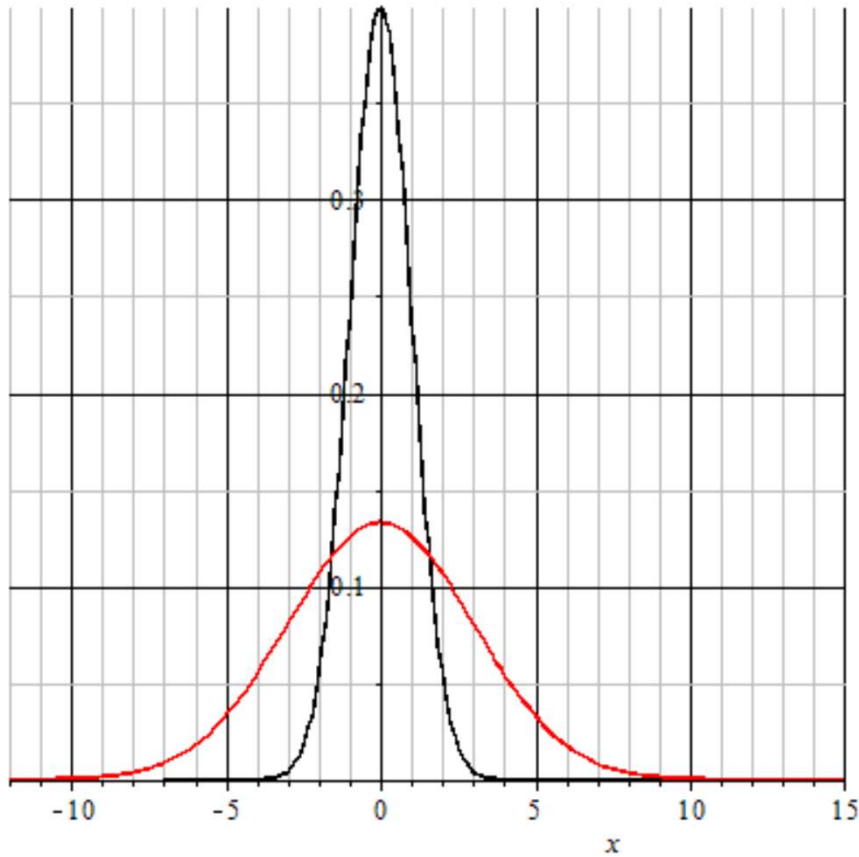
Graph of normal distribution density function (32) is represented on Fig.





The black graph corresponds to the expected value μ_x equal to zero, and the red to the expected value μ_x equal to five.

The expected value corresponds to the maximum of the density function.



The black graph corresponds to sigma equal to one, and the red sigma equal to 3.

The height and width of the graph depends on sigma: the smaller the sigma, the higher and narrower the graph.

Normal distribution is the most often meeting in various casual natural phenomena. So, mistakes of performance of commands by automated device, mistakes of conclusion of spacecraft in the given point of space, mistakes of parameters of computer systems, etc. in the most cases have normal or close to normal distribution.

Moreover, random variables formed by summing of great quantity of random addendums, are distributed practically according to normal law.

Example 2.9

A continuous random variable has a normal distribution with $\mu_X = 0$, $\sigma = 1$.

- a) With MAPLE find the probability in the segment $[-\infty, \mu_X - \sigma]$
- b) With MAPLE find the probability in the segment $[\mu_X - \sigma, \mu_X + \sigma]$.
- c) With MAPLE find the probability in the segment $[\mu_X + \sigma, +\infty]$.
- d) Find $P_a + P_b + P_c$

SOLUTION

$$P(\mu_X - \sigma < X < \mu_X + \sigma) = \int_{\mu_X - \sigma}^{\mu_X + \sigma} f(x) dx$$

$$P_b = P(\mu_X - \sigma < X < \mu_X + \sigma) = 0.683$$

This is the one sigma rule.

$$P(-\infty < X < \mu_X - \sigma) = \int_{-\infty}^{\mu_X - \sigma} f(x) dx$$

$$P_a = P(-\infty < X < \mu_X - \sigma) = 0.159$$

Example 2.10

A continuous random variable has a normal distribution with $\mu_x = 0$, $\sigma = 1$.

- a) With MAPLE find the probability in the segment $[-\infty, \mu_x - 3\sigma]$
- b) With MAPLE find the probability in the segment $[\mu_x - 3\sigma, \mu_x + 3\sigma]$.
- c) With MAPLE find the probability in the segment $[\mu_x + 3\sigma, +\infty]$.
- d) Find $P_a + P_b + P_c$